Soak the seeds of chick pea or gram in water for 2 – 3 hours. Drain the water and place the seeds in a moist cotton cloth for 2 – 3 days. Observe for germination. Select some sprouted seeds, observe under a dissection microscope and record the parts.

Diagnostic Features

- Seeds of gram have two cotyledons and an embryonal axis.
- Each seed is covered by two seed coats (a) Testa – outer coat and (b) Tegmen – inner coat.
- The embryonal axis consists of radicle and plumule.
- The portion of the embryonal axis above the level of cotyledons is called epicotyl. It terminates into the plumule.
- The portion of the embryonal axis below the level of cotyledons is called hypocotyl. It terminates into the radicle or root tip.

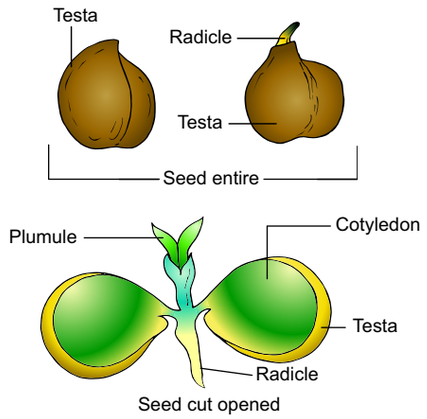


Figure 6: Dicot seed – Gram (*Cicer*)

Exercise 7: Dispersal of seeds by various agents

Aim: To study and understand the agents that help in the dispersal of fruits and seeds.

Principle: The dissemination of seeds and fruits to various distances from the parent plant is called seed and fruit dispersal . It takes place with the help of ecological factors such as wind, water and animals.

Requirements: Fruits of *Tridax*, Coconut and *Achyranthes*, knife, forceps, petridish, hand-lens etc.

7 A. Dispersal by wind – Anemochory (Example: *Tridax*)

Diagnostic Features

- Fruits are light so that wind may carry them away.
- Fruits are minute, very small and with inflated covering.
- Fruits have feathery appendages (pappus) which greatly increase their buoyancy to disperse in high altitudes.

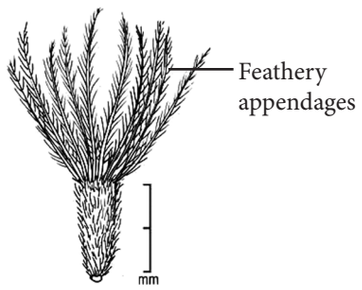


Figure 7a: Fruit of *Tridax*

7 B. Dispersal by water – Hydrochory (Example: Coconut)

Diagnostic Features

- Fruits have outer coats that are modified to enable them to float.
- The mesocarp of coconut is fibrous, which is easily carried away by water currents.



Figure 7b: Coconut

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